

Project: Grant Finder

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Introduction

Finding suitable funding opportunities can often be a difficult and time-consuming process. Students and researchers must search across multiple databases, websites, and funding platforms, each with its own structure, eligibility requirements, and search interface. As a result, identifying grants that match a person’s academic background and research interests can require significant effort.

One possible way to simplify this process is through Retrieval-Augmented Generation (RAG). By combining information retrieval with large language models, such systems can search large collections of documents, identify relevant opportunities, and present results in a more accessible and personalized way.

This paper presents the design and implementation of the Grant Finder, a RAG-based recommendation system that helps users to discover funding opportunities relevant to their profile. The system analyzes a user’s CV and research interests, uses retrieved information from grant databases, and generates personalized recommendations together with explanations of why particular opportunities may be suitable for a specific user.

Since the system processes personal information and can influence users’ decisions, this work also discusses ethical aspects of AI-based recommendation systems, including possible risks, limitations, and future improvements from both technical and ethical perspectives.

1 Background

1.1 Retrieval-Augmented Generation (RAG)

One of the main limitations of Large Language Models (LLMs) is that they generate responses based on knowledge learned during training. As a result, they may provide outdated information, hallucinate facts, or fail to incorporate newly available data.

Retrieval-Augmented Generation (RAG) was introduced to address this limitation (Gade et al.). Instead of answering a question using only the model’s internal knowledge, a RAG system first searches for relevant information in an external knowledge base. The retrieved information is then provided to the language model as additional context for generating a response.

A typical RAG pipeline (Figure 1) consists of document collection, text splitting, embedding generation, retrieval, and answer generation (Fan et al.). By using external sources, RAG systems can improve grounding and provide more transparent answers.

Simple RAG

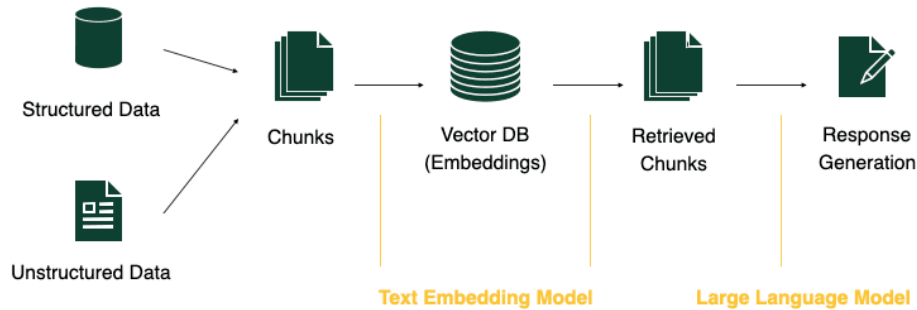


Figure 1: RAG structure

1.2 RAG for Recommendation Systems

Recommendation systems aim to help users discover relevant information from large collections of available options. Traditional recommendation approaches often rely on user history, predefined categories, or similarity-based matching. However, in research funding, users may not know which opportunities exist or which keywords describe their interests.

RAG can improve recommendation systems by combining semantic retrieval with natural language generation. The retrieval component searches a large knowledge base for relevant items, while the language model can analyze results and explain why they match the user's needs.

This approach allows the system to compare a researcher's background with grant opportunities and make personalized suggestions. In addition, retrieved sources can be shown together with recommendations, improving transparency and allowing users to verify the provided information.

2 The Grant Finder System

2.1 Project Objectives

The goal of the Grant Finder is to help students find relevant grants and funding opportunities. By analyzing a student's academic background, skills, and research interests, the system generates personalized recommendations and highlights opportunities that may be a good match for the user's profile.

Unlike traditional search engines, which often require users to manually browse multiple websites and databases, the Grant Finder combines reliable information with natural

language generation. This allows the system not only to identify relevant opportunities but also to explain why a particular grant may be suitable for a specific user.

2.2 System Architecture

At present, the system uses the European Union Funding and Tenders (EU F&T) API as its main data source (European Commission). Grant information is collected, preprocessed, and stored in a structured JSON format. Each record contains information such as the grant title, description, funding amount, deadlines, eligibility requirements, and source information.

To improve retrieval quality, grant descriptions are divided into chunks before indexing. Each chunk has a length of 1750 symbols and contains a main text segment (1000 symbols) together with general grant context (750 symbols) and metadata (Figure 3b). This approach helps to preserve important information that may otherwise be lost (Kiss et al.; Byte Vagabond).

The chunks are converted into vector embeddings using **all-MiniLM-L6-v2** sentence-transformer model and stored in a Chroma vector database. This allows the system to perform semantic search instead of only keyword matching.

To improve retrieval quality, the system also uses an LLM-based query expansion step that enriches the query with related concepts and funding-specific terms. Retrieved chunks are ranked using Maximal Marginal Relevance (MMR) before being provided to the language model as supporting context.

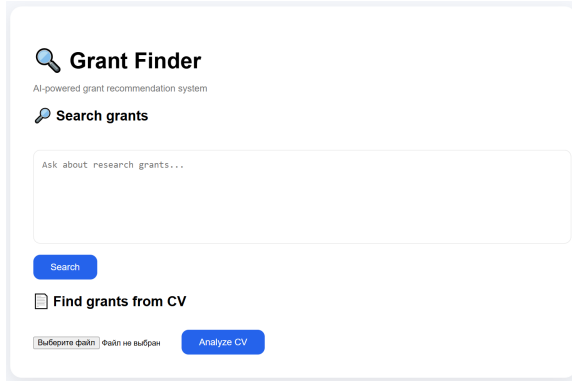


Figure 2: Grant Finder architecture

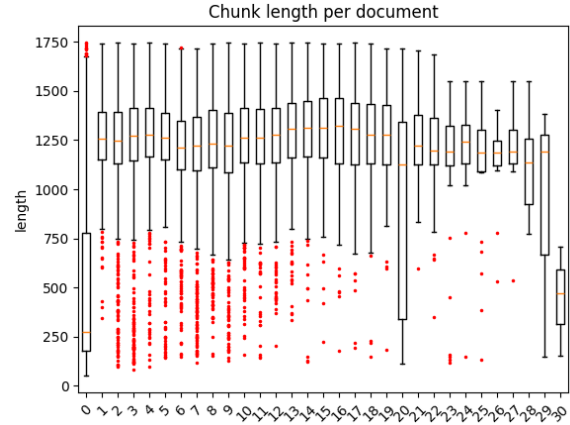
2.3 Recommendation Generation

The system supports two recommendation modes. In the first mode, users can ask general questions about funding opportunities. The expanded query (Gade et al.) is used to retrieve relevant grant information from the vector database, and the retrieved context is provided to the **Qwen3.5-0.8B** language model, which generates the final response.

In the second mode, users upload a CV in PDF format. The system extracts information about education, skills, experience, and research interests to create a user profile. This profile is then used to retrieve relevant funding opportunities from the database.



(a) User Interface



(b) Chunk length distribution

2.4 User Interface

The frontend is implemented using Flask and provides a simple web interface. Users can either enter a text query or upload a CV for personalized recommendations. Figure 3a shows the current version of the interface.

The system was designed around a retrieval-based architecture rather than relying only on language model generation. This approach provides access to current grant information and allows recommendations to be grounded in external sources. However, as discussed in the next section, the retrieval layer also introduces its own ethical challenges and potential sources of risks.

2.5 Example Recommendation

To demonstrate the functionality of the Grant Finder, two examples are presented: a natural language query (Figure 4b) and a CV-based recommendation (Figure 4a).

Answer

- Towards Artificial General Intelligence (AGI) for healthcare**
2. Match: The profile explicitly lists "medical diagnostics" and "semiconductor defect detection" as core competencies, which aligns directly with the "Biomedical software" and "Health sciences" keywords in the grant summary. The grant emphasizes "Personalised medicine" and "Data protection and privacy," addressing the specific barriers mentioned in the profile regarding "scalability" and "regulatory frameworks."
3. Funding: 2,900,000 – 12,000,000 EUR
4. Deadline: 2027-09-22
5. Source: [1]
- Automated Scientific Discovery (RAISE pilot)**
7. Match: The profile includes "data engineering," "pipeline automation," and "ablation studies," which are explicitly mentioned as key requirements in the RAISE grant summary. The grant focuses on "Assistive and interactive safe AI-managed robotic systems" and "Collaborative experimentation," directly addressing the need for scalable automation and networked AI systems described in the profile.
8. Funding: 3,000,000 – 10,000,000 EUR
9. Deadline: 2027-02-02
10. Source: [2]
- EU Frontier AI Initiative: Developing frontier AI solutions that are safe and computationally efficient within Apply AI (RIA)**
12. Match: The profile highlights "Transformer architecture," "PyTorch," and "Neural networks," which are central to the "AI-Driven Robotics" and "Frontier AI" initiatives. The grant emphasizes "Agentic AI frameworks" and "Trust and safe adoption," addressing the "multi agent systems" and "trust" challenges raised in the profile regarding "scalability" and "regulatory frameworks."
13. Funding: 1,500,000 – 44,000,000 EUR
14. Deadline: 2027-03-18
15. Source: [4]
- Apply AI: Piloting AI-based image screening in medical centres**
17. Match: The profile includes "Computer Vision," "Object Detection," and "YOLOv5," which are explicitly mentioned in the grant summary as key capabilities. The grant focuses on "AI-Driven

Sources

- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-HLTH-2027-03-TOCL-08>
- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-RAISE-2027-01-01>
- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/DIGITAL-2026-SKILLS-10-EDITS-CH>
- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-CL4-2027-04-SERV-01-02>

Search grants

top 3 AI in healthcare grants

Find grants from CV

Answer

Based on the provided context, here are the top 3 AI in healthcare grants:

- Towards Artificial General Intelligence (AGI) for healthcare**
• Funding: 2,900,000 – 12,000,000 EUR
• Deadline: 2027-09-22
• Source: [8]
- Clinical research by Comprehensive Cancer Infrastructures for the benefit of patients with common cancers**
• Funding: 3,000,000 – 10,000,000 EUR
• Deadline: 2027-09-21
• Source: [4]
- Personalised approaches to reduce risks from Adverse Drug Reactions due to administration of multiple medications**
• Funding: 3,000,000 – 11,000,000 EUR
• Deadline: 2027-04-13
• Source: [5]

Sources

- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-HLTH-2027-02-DISEASE-01-two-stage>
- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-HLTH-2027-02-DISEASE-14-two-stage>
- <https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/topic-details/HORIZON-INFRA-2027-SERV-01-02>

(a) Example 1

(b) Example 2

3 Ethical Analysis of the Grant Finder

3.1 Reliability of Recommendations

A main ethical challenge of the Grant Finder is the possibility of incorrect recommendations. Although the system uses RAG to ground answers in retrieved grant information, errors are still possible. They can happen when relevant documents are not retrieved, when the database contains outdated information (Gade et al.), or when the language model misunderstands the retrieved context.

In practice, these errors could result in incorrect deadlines, funding amounts, or eligibility requirements being presented to users. Since researchers may rely on this information when planning applications, even a single incorrect recommendation can influence important academic or career decision.

The risks associated with AI-generated recommendations are illustrated by the widely discussed Air Canada case. In that case, a customer relied on information provided by a chatbot regarding a bereavement fare discount. The information was incorrect, and the court ultimately held Air Canada responsible for the chatbot's output (Moffatt and Rivers). The case demonstrates that users often treat AI-generated information as trustworthy, even when the system is intended only as an advisory tool.

To reduce this risk, the system should provide source references for generated answers and clearly indicate when information is missing. The Grant Finder should support

decision-making, but users should always verify important details using official grant sources.

3.2 Information Visibility and Retrieval Power

Another important challenge comes from the retrieval process itself. Before generating an answer, the system selects only a limited number of relevant grant documents from the vector database. Therefore, the retrieval component directly affects which opportunities are shown to users. A suitable grant may not appear in the recommendation because of limitations in embeddings, chunking strategy, ranking methods, or incomplete data. This creates a visibility problem: some opportunities may become hidden even though they could be relevant.

The current implementation attempts to improve retrieval quality through semantic embeddings, query expansion, and Maximal Marginal Relevance (MMR) ranking. Future work may include reranking models, improved chunking strategies, using subqueries, and additional data sources to reduce the risk of relevant opportunities being overlooked (Gade et al.).

3.3 Accountability and Human Oversight

The Grant Finder is designed as an advisory tool, not an autonomous decision-maker. However, its recommendations may still influence user behavior. Students may use the system to identify opportunities or prioritize their applications. As a result, incorrect recommendations can affect important educational and career-related decisions.

Therefore, the responsibility is shared between the data sources, retrieval methods, language model, developers, and users. For this reason, transparency is an important design principle for the system. Recommendations should be accompanied by source references and the retrieval process should remain visible to users whenever possible. Logging and traceability mechanisms could also improve accountability by making it possible to reconstruct how a recommendation was generated and which sources influenced the output.

4 Future Development

Several improvements are planned for future versions of the Grant Finder. The system could be extended with additional grant databases and APIs together with deduplication mechanisms to improve coverage and reduce duplicate recommendations. Retrieval quality could also be improved with the use of hybrid retrieval approaches that combine semantic and keyword-based search, reranking models, and alternative chunking strategies.

Another important direction is improving the reliability of generated recommendations. Future versions may incorporate evaluation and validation techniques such as

self-reflection, LLM-as-a-judge, probabilistic confidence estimation, and combined approaches similar to Trustworthy Language Models (TLM)(Goh et al.).

The quality of recommendations could also be improved by using stronger language models with larger context window. Finally, transparency and accountability could be enhanced through automatic source attribution, traceability mechanisms, and detailed logging of the retrieval and generation process (Fan et al.).

5 Discussion

The project demonstrates that a functional RAG-based recommendation system can be developed with relatively limited resources in a university environment. The implementation demonstrates the feasibility of applying RAG to grant recommendation tasks by combining semantic retrieval with natural language generation.

The main advantage of the Grant Finder is not only the use of a language model itself, but the combination of multiple components: semantic retrieval, external knowledge integration, source traceability, and profile-based matching. This allows the system to provide more personalized recommendations compared to traditional search-based approaches. Table 1 summarizes the main differences between the functional capabilities of proposed system and commonly used alternatives.

Table 1: Comparison of Grant Finder with existing solutions

System	Search method	Personalization	Natural language	Source transparency	CV matching
EU Funding & Tenders Portal	Keyword/filter search	Limited	Limited	Yes	No
Google Search	Keyword + ranking	No	Partial	Depends on source	No
Generic LLM chatbot	LLM knowledge	Possible	Yes	Usually limited	Possible
Grant Finder	Semantic retrieval + RAG	Yes	Yes	Yes	Yes

Compared with the EU Funding & Tenders Portal, the proposed system provides additional personalization by allowing users to search using natural language queries and CV-based profiles. Traditional search platforms require users to manually define filters and often provide limited contextualization.

At the same time, the project highlights several challenges discussed throughout this paper. Recommendation quality depends not only on the language model but also on the quality of source data, retrieval methods, chunking strategies, and ranking mechanisms.

As a result, improving model performance alone is not sufficient to guarantee reliable recommendations. Building a truly trustworthy and practically useful system requires improvements across the entire pipeline, including data integration, retrieval quality, and user interaction design.

The current implementation has several limitations. It relies on a single API source, uses a relatively small language model, and has not been evaluated using retrieval and generation quality metrics. In addition, the current prototype provides only basic source attribution, and the user interface could be further improved.

6 Conclusion

The Grant Finder demonstrates the potential of RAG-based systems for personalized research funding search by combining external information retrieval with natural language generation. The developed prototype shows that this approach can make the process of discovering suitable opportunities more efficient and user-friendly.

At the same time, the work highlights that the reliability of RAG-based recommendation systems depends on the entire pipeline rather than only on the language model. Data quality, retrieval methods, ranking strategies, and transparency mechanisms all play an important role in determining whether generated recommendations are useful and trustworthy.

Future development of the Grant Finder should focus on expanding data sources, introducing systematic evaluation methods, and improving recommendation traceability. RAG-based systems should be designed to support users by providing relevant and verifiable information while keeping their limitations transparent.

Works Cited

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A AI Use and Research Transparency Statement

During the work on the paper, AI tools were used as a research and writing support tool. In particular, ChatGPT-5 model (July 2026 version) was used for:

- writing code snippets
- explanation of the usage of libraries
- assistance with error resolving
- designing frontend components
- discussing the structure of the paper
- improving grammar and readability of the certain paragraphs
- generating alternative formulations of arguments for comparison
- provision of feedback on the clarity and coherence of the paper

Examples of prompts:

- How to read pdf files? which modules can you suggest for the project?
- How to make text wrap around an image in latex
- Suggest a structure for a seminar paper about a RAG system that I developed. The paper should describe its objectives, architecture, implementation, and ethical challenges. I need to explain the system and connect it to the ethical questions discussed in my presentation. The system is functional, but some of the responsible AI mechanisms discussed in the paper have not yet been implemented and are planned as future work. Future versions of the system will be deployed on university servers and integrated with additional API.
- Review this section of my paper and propose improvements to clarity, structure, and writing style without changing my logic or the main argument.

A.1 Author’s Independent Contribution

AI generated content was not used as a source of factual information, all factual claims, statistics and case descriptions were taken from the supplied sources and verified.

The overall system design, architectural decisions, data entries design, chunking strategy, retrieval pipeline, model selection, and integration of system components were developed independently by the author. AI code suggestions were reviewed, modified, and integrated into the project when appropriate.

Additionally, the analysis and interpretation of results presented in the paper were not AI-generated. The AI-tools did not determine the direction of analysis, formulation of final arguments and conclusions made in the study.